



CivicEngine

AI-Powered Multimodal Indian Municipal Grievance Resolution Center

Google Code for Community 2026

India Civic Tech Hackathon Submission

Empowering Citizens & Wards

The Civic Grievance Bottleneck

- ✦ **Language Barriers:** Traditional portals require text entry in standard English/Hindi, excluding millions of regional-language speakers.
- ✦ **Mismatched Ward Routing:** General citizen complaints route to centralized queues instead of local ward officers, leading to massive SLA delays.
- ✦ **Redundant Duplicates:** Dozens of citizens report the same pothole or heap, causing administrative noise and database bloat.
- ✦ **Lack of Real-world Action:** Delayed complaints rotting in lists leave citizens powerless without a legal follow-up receipt.

The Citizen Slop

Clunky text forms, manual drop-down ward selections, and no voice-based assistance lead to drop-offs.

Operational Overhead

Ward officers are flooded with identical duplicate reports, requiring manual review and sorting.

The CivicEngine Framework

Bridging community needs with legal power and advanced AI. An end-to-end PWA & Android app designed to process civic issues seamlessly at the ward level.



Multimodal Voice Intake

AI Live Operators converse in Hindi, Tamil, Telugu, Kannada, or English. Transcribes, translates, and filters spam instantly.



Ward-Level Routing

GPS locations map to exact municipal boundaries via ISRO Bhuvan/OSM. High-priority queues auto-calculate urgencies.



Legal RTI Empowerment

Generates a pre-filled RTI Act 2005 receipt for delayed cases, enabling citizens to legally question PIOs.

Gemini Live Voice Assistant

- ✦ **Background Thread AudioWorklet:** Migrated from deprecated thread-blocking ScriptProcessors to AudioWorklets, ensuring zero microphone lag on mobile.
- ✦ **Speech Modality (Zephyr):** Direct connection using Gemini's live audio interface. Speaks like an empathetic emergency operator.
- ✦ **Direct Tool Integration:** Gemini queries details and calls the `submit_report` tool function seamlessly once it has category, severity, and location.
- ✦ **Full i18n Core:** Auto-transcribes from 5 languages, outputting clean English summaries for officers.

LIVE DEMO SESSION CONFIG

"Converse in Citizen's Local Language..."

"Namaste/Vanakkam, I am your CivicEngine Operator. Please tell me what issue you are facing and your nearby landmarks..."

MODEL: `gemini-2.0-flash-exp` (WebSockets)

Precise Location Routing

- ◆ **Local Boundaries cache:** Point-in-polygon mapping of coordinates against DataMeet shape files for India.
- ◆ **OpenStreetMap Overpass Boundary Resolution:** Fallback query resolving administrative boundaries at `admin_level=10` (Ward level in Indian Municipalities).
- ◆ **Google Maps Geocoding API:** Last-resort fallback for neighborhoods and sublocalities.
- ◆ **Geohash Indexing:** Real-time Firestore geohashes for high-performance spatial querying.

Ward Boundary Resolver API

POST /api/detect-ward

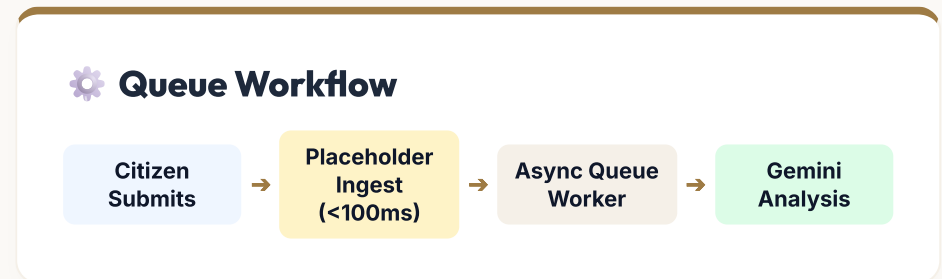
Input: { latitude: 12.971, longitude: 77.594 }

Output:

```
{ ward: "Sampangiram Nagar", wardId: "osm_relation_12345",  
  municipality: "BBMP", state: "Karnataka" }
```

Scalable Async AI Queue Worker

- ◆ **Instantly Non-Blocking:** Citizens receive a tracking number in <100ms. The heavy AI analysis runs asynchronously in a worker thread.
- ◆ **Rate Limit Defense:** Handles Gemini 429 Quota Exceeded using backoffs, preventing request thread starvation.
- ◆ **Regex Fallback Triager:** If API limits are exhausted, fallbacks auto-categorize issues, keeping the platform 100% operational.
- ◆ **Background Asset Sync:** Stores and parses media buffers directly from GCS/local data URIs.



Semantic Duplicate Prevention

- ◆ **Ward Pre-Filtering:** Fetches active issues in the same category & ward first, avoiding expensive global database queries.
- ◆ **AI Similarity Checker:** Gemini reviews active complaints and compares location details to detect matches.
- ◆ **Auto-Upvote Conversion:** If matched, the new report is merged. The original issue is automatically upvoted and re-ranked.
- ◆ **Real-time UX Redirect:** Citizen gets an overlay:
*"Duplicate report resolved! Upvoted existing case
Reference: CE/..."*

✓ Operational Efficiency

Instead of 50 separate tickets for one open manhole, authorities get **one ticket with 50 upvotes**, pushing it instantly to the top of the urgency queue.

SAVES 90% TRIAGE TIME

RTI Act 2005 Legal Integration

- ◆ **Citizen Legal Power:** Pre-filled RTI application drafts are generated for overdue or unresolved complaints.
- ◆ **Section 6(1) Pre-formatting:** Automates standard Indian RTI structures, referencing Public Information Officer (PIO) offices.
- ◆ **Pre-populated Case Metrics:** Inserts GPS coordinates, category, reported date, tracking ID, and delay timeline logs.
- ◆ **Direct Receipt Download:** Compiles a ready-to-print text receipt document in one click.

LEGAL APPEAL DRAFT

"To, The Public Information Officer (PIO)...
Subject: Request for Information under Sec 6(1) of RTI Act 2005.
Particulars of grievance: ID CE/2026/07/00042.
Please provide: (1) Files notes and logs on resolution action. (2)
Budget allocated for this category in Ward 4."

Android Native App Wrapper

- ✦ **Capacitor Native Integration:** The entire React PWA is packaged inside a native Android Studio Gradle project (located in /android).
- ✦ **Offline Synchronization:** Built web assets sync straight to local assets, allowing instant app boots.
- ✦ **Responsive Sandalwood Theme:** Premium styling following WCAG 2.1 AA accessible contrast guides.
- ✦ **Real-time Leaflet Maps:** Fully optimized touch gestures for locating nearby municipal issues and boundaries.

Android APK

Generated project ready for release build, launching via Capacitor Native Bridges.

Offline Ready

Fully functional PWA manifest and service workers cache local assets and queue drafts.

Google Cloud & Stack Overview

A robust, modern full-stack architecture built to scale to lakhs of users.

Frontend & Mobile

- **Core:** React, TypeScript, Vite, Tailwind CSS
- **Mapping:** Leaflet JS, OpenStreetMap Nominatim
- **Voice:** Web Audio API, AudioWorklets
- **Mobile Wrapper:** Capacitor SDK (Android Platform Gradle Project)

Backend & AI (GCP asia-south1)

- **Server:** Node.js, Express, tsx (WebSocket Server)
- **AI Engine:** @google/genai SDK (Gemini 2.5 Flash, Gemini 2.0 Live)
- **Database:** Google Cloud Firestore (asia-south1 region)
- **Storage & Auth:** Firebase Client/Admin Auth, Google Cloud Storage

Social Impact & Governance

IN Regional Inclusion

Empowers rural and urban citizens to file reports verbally in native dialects.
Overcomes reading and writing constraints.

Transparency Logs

All status updates and department reallocations log public audit histories, preventing official neglect.

Public Budget Check

Enables citizens to inspect allocated infrastructure budgets when filing complaints, promoting accountability.

Roadmap & Future Vision

- ✦ **Municipal Integration:** Standard pilots with BBMP (Bengaluru), GCC (Chennai), and BMC (Mumbai) public grievance cells.
- ✦ **ISRO Bhuvan Integration:** Transitioning boundary resolution to India's satellite mapping layers.
- ✦ **RTI Digitization:** Direct API connection to central/state RTI portals for single-click official online filing.
- ✦ **Community UPI Rewards:** Crowdsourced verify-and-cleanup incentives with mock micro-transactions.

IN

Code for Community

"Building technology that bridges legal rights, native communication, and state-of-the-art AI to transform Indian civic administration."